

Table 2: Examples of problems from each pragmatic phenomenon that PRAGMEGA includes. **Bold** indicates the correct answers. In the actual dataset, the order of choices is randomly shuffled. We adapt the table from examples shown in (Hu et al., 2023).

Phenomenon	Questions	Options
Deceits	Henry is sitting at his desk and watching TV, and reluctantly switches off the TV with the remote control and picks up a textbook. Shortly after, his mother comes in the room and asks, “What have you been doing up here?” Henry responds: “Reading.” Why has Henry responded in such a way?	1. He does not want to get into trouble for not studying. 2. He has been reading for some time. 3. He does not want to offend his mom by not reading the books that she gave him. 4. He wants his mom to believe that he has been watching TV.
Metaphor	Andrew and Bob were discussing the investment company where Andrew works. Bob said: “The investors are squirrels collecting nuts.” What does Bob mean?	1. They buy stocks hoping for future profit. 2. Squirrels were hired to work in the company. 3. The investors dress and eat well. 4. Bob is allergic to nuts. 5. The investors enjoy picking nuts as squirrels do.
Indirect Speech	Nate is about to leave the house. His wife points at a full bag of garbage and asks: “Are you going out?” What might she be trying to convey?	1. She wants Nate to take the garbage out. 2. She wants to know Nate’s plans. 3. She wants Nate to bring his friends over. 4. She wants Nate to spend more time with the family.
Irony	It is a holiday. Stefan and Kim are sitting in the backseat of the car. They are fighting all the time. Their father says: “Oh, it is so pleasant here.” What did the father want to convey?	1. He does not want to listen to his kids’ arguments. 2. He enjoys listening to his kids fighting. 3. AC gives them some needed cool. 4. He remembers about his wife’s birthday.
Maxims	Leslie and Jane are chatting at a coffee shop. Leslie asks, “Who was that man that I saw you with last night?” Jane responds, “The latte is unbelievable here.” Why has Jane responded like this?	1. She does not want to discuss the topic that Leslie has raised. 2. She thinks that it is the best latte in the town. 3. The man who Leslie saw makes unbelievable lattes. 4. A coffee break is not a good time to discuss men.

Table 3: Prompts used in the methods compared in this study.

Method	Prompt
Simple (Baseline-1)	Write ONLY the option number of your final answer and its contents in the format like: [Answer] 2) hogehoge is hogehoge. Any additional output beyond this penalized.
Chain-of-Thought (Baseline-2)	Firstly, think step-by-step and write down your process of thinking. After that, select your final answer. Your final answer should be in the format like: [Answer] 2) hogehoge is hogehoge.
Gricean Prompting (Proposed-1)	Let's think in line with the Gricean theory. In Grice's framework, hearers arrive at the implied meanings (or "implicatures") through an inferential process guided by the Cooperative Principle and its associated conversational maxims (Quantity, Quality, Relation, and Manner). Specifically: Cooperative Principle: The assumption that speakers and hearers are cooperating with one another to communicate effectively. Conversational Maxims: Quantity: Be as informative as required (but not overly so). Quality: Do not say what you believe to be false or lack evidence for. Relation (Relevance): Be relevant. Manner: Be clear, avoid ambiguity and obscurity. When a hearer detects a potential mismatch between what is said (literally) and one of the maxims, they hypothesize a conversational implicature—that the speaker must mean something more or different than the literal meaning. The hearer then uses context, background knowledge, and reasoning about the speaker's intent and adherence to the maxims to infer the intended meaning. Write down your thinking process in line with Gricean theory and ultimately decide on the final answer. Your final answer should be in the format like: [Answer] 2) hogehoge is hogehoge.
Relevance Theory Prompting (Proposed-2)	Let's think in line with the Relevance theory. According to Relevance Theory, the interpretation of utterance implicatures proceeds through the following processes, where the balance between cognitive effects and processing effort plays a crucial role. — 1. Starting Point of Utterance Interpretation: - The linguistic meaning (logical form) of an utterance is merely a "clue" to the interpretation intended by the speaker. - The listener must infer the speaker's intended meaning behind the utterance using this linguistic clue as a basis. ... 6. Interaction Between Explicit Meaning and Implicature: - Explicit meaning (the overt content of the utterance) and implicature (implied content) influence each other during processing. - This interaction forms the overall interpretation of the utterance. In Relevance Theory, "optimal relevance" is achieved when an utterance delivers "cognitive effects worth the processing effort" to the listener. Thus, the balance between cognitive effects and processing effort is consistently emphasized in utterance interpretation. By seeking interpretations that maximize effects with minimal effort, listeners achieve efficient understanding of utterances. Write down your thinking process in line with Relevance theory and ultimately decide on the final answer. Your final answer should be in the format like: [Answer] 2) hogehoge is hogehoge.
Short Gricean Prompting (Short-1)	Let's think in line with the Gricean theory. Write down your thinking process in line with Gricean theory and ultimately decide on the final answer. Your final answer should be in the format like: [Answer] 2) hogehoge is hogehoge.
Short Relevance Prompting (Short-2)	Let's think in line with the Relevance theory. Write down your thinking process in line with Relevance theory and ultimately decide on the final answer. Your final answer should be in the format like: [Answer] 2) hogehoge is hogehoge.

Table 4: Prompts used in the additional experiments to examine the first concern.

Method	Prompt
xbar	Let' s think in line with the X-bar theory. Write down your thinking process in line with X-bar theory and ultimately decide on the final answer. Your final answer should be in the format like: [Answer] 2) hogehoge is hogehoge.
complexity	Let' s think in line with the computational complexity theory. Write down your thinking process in line with computational complexity theory and ultimately decide on the final answer. Your final answer should be in the format like: [Answer] 2) hogehoge is hogehoge.
graph	Let' s think in line with the graph theory. Write down your thinking process in line with the graph theory and ultimately decide on the final answer. Your final answer should be in the format like: [Answer] 2) hogehoge is hogehoge.

E Hyperparameters

We set temperature=0.8, max_new_tokens=1500, repetition_penalty=1.2, do_sample=True.

F Computation Details

The parameter counts for each open model we used are as follows:

- Llama-3.1-8B-Instruct: 8B
- Qwen2.5-7B-instruct: 7B
- Qwen2.5-14B-instruct: 14B
- phi-4: 14B

We used NVIDIA RTX™ A6000, and our experiments took around 200 GPU hours. All models were quantized to 8-bit before performing inference using vLLM (Kwon et al., 2023). The experiments were conducted on March 2, 2025.

G Exact Experimental Results

In our main experiments, the exact accuracy achieved by each model is shown in 7.

In addition, the detailed results for each method used in the additional experiments are presented in Table 8.

H Correlation Analyses between Accuracy and Input/output Length

The results of the analysis on the relationship between Accuracy scores and the length of Input to the model and the length of Output from the model are shown in Table 5 and 6, respectively.

I Exact Results for Each Phenomena

The exact experimental results aggregated for each phenomenon in PRAGMEGA are shown in Table 9, 10, 11, 12, 13, and 14.

J Examples of Each Error Pattern

Table 15 shows specific examples of each error pattern by GPT-4o presented in Sec. 4.2.3.